

Unified Recruitment Mediation Platform – End-to-End Model Explanation

1. Introduction

This project acts as a mediator between recruiters and job seekers. The platform automates initial screening, assessment, and ranking of candidates using self-designed algorithms, ensuring fairness, transparency, and realism aligned with modern recruitment practices.

2. System Actors

- Recruiter (Company HR)
- Candidate (Job Seeker)
- System (Recruitment Mediation Platform)

3. Recruiter Workflow

1. Recruiter logs into the recruiter portal.
2. Recruiter creates a job posting with job title, description, required skills, experience, salary, and location.
3. Recruiter sets internal options (not visible to candidates):
 - Coding assessment required (Yes/No)
 - Skill importance (Mandatory / Optional)
 - Assessment mode (Platform-generated or Recruiter-provided questions)
 - Scoring weight preferences (Resume, Assessment, Interview)
4. Job is published and becomes visible to candidates.

4. Candidate Workflow

1. Candidate logs into candidate portal.
2. Candidate browses available job postings.
3. Candidate views job details and applies for a job.
4. Candidate submits personal details, resume, and basic screening information.

5. Resume Parsing & Skill Matching (Gate-1)

1. System extracts text from resume.
2. Custom skill extraction algorithm identifies skills based on keyword frequency and context.
3. Extracted skills are matched with job-required skills.
4. Mandatory skills must be present; optional skills add score weight.
5. If match criteria fails, candidate is rejected politely. Otherwise, candidate proceeds.

6. Assessment Generation & Evaluation (Gate-2)

1. System checks assessment mode selected by recruiter.
2. If platform-generated, assessment is created based on matched skills and difficulty levels.
3. If recruiter-provided, recruiter's own questions are used.
4. Coding assessment is included only if recruiter enabled it.
5. Candidate completes MCQ and coding assessment (if applicable).
6. System evaluates responses and generates assessment score.

7. AI Interview Module

1. Candidate attends a short 1–2 minute interview.
2. Fixed behavioral and communication prompts are asked.
3. Responses are captured and scored.

8. Unified Scoring & Ranking

1. System calculates individual scores:
 - Resume skill match score
 - Assessment score
 - Coding score (if applicable)
 - Interview score
2. Recruiter-defined weights are applied.
3. Bias mitigation rules ensure no single factor dominates unfairly.
4. Final unified score is generated and candidates are ranked.

9. Recruiter Decision & Handoff

1. Recruiter views ranked candidate list with score breakdown.
2. Recruiter shortlists candidates based on cutoff criteria.
3. Job is closed.
4. Shortlisted candidates receive interview or selection emails.
5. Rejected candidates receive polite notifications.
6. Platform exits the recruitment process and enables direct recruiter–candidate communication.

10. Conclusion

The platform provides an end-to-end, realistic, and fair recruitment workflow using self-implemented algorithms and structured decision-making. It mirrors modern recruitment systems while remaining academically sound and explainable.